



# Yakeen Plus (2025)

## Short Practice Test - 09

**DURATION: 60 Minutes**

**DATE: 17/11/2024**

**M. MARKS: 192**

### Topics Covered

|                   |                                                                                                                                                                                                 |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Physics:</b>   | Dual Nature of Radiation and Matter, Atoms, Nuclei, Semiconductor Electronics: Materials, Devices and Simple Circuits                                                                           |
| <b>Chemistry:</b> | Practical Organic Chemistry, Coordination Compound, The d and f-Block Elements                                                                                                                  |
| <b>Biology:</b>   | <b>(Botany)</b> : Organisms and Population, Ecosystem, Biodiversity and Conservation<br><b>(Zoology)</b> : Biotechnology and its Applications, Evolution, Biotechnology: Principles & Processes |

#### General Instructions:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **60 minutes** duration.
3. The test booklet consists of **48** questions. The maximum marks are **192**.
4. All questions are compulsory.
5. There is only one correct response for each question.
6. Each correct answer will give **4** marks while **1** Mark will be deducted for a wrong MCQ response.
7. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.
9. **Do not fold or make any stray mark on the Answer Sheet (OMR).**  
**OMR Instructions:**
  1. Use blue/black dark ballpoint pens.
  2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
  3. Never use pencils to mark your answers.
  4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
  5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
  6. Multiple markings will be treated as invalid responses.
  7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

**Name of the Student (In CAPITALS) :** \_\_\_\_\_

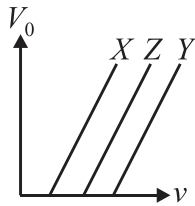
**Roll Number :** \_\_\_\_\_

**OMR Bar Code Number :** \_\_\_\_\_

**Candidate's Signature :** \_\_\_\_\_ **Invigilator's Signature** \_\_\_\_\_

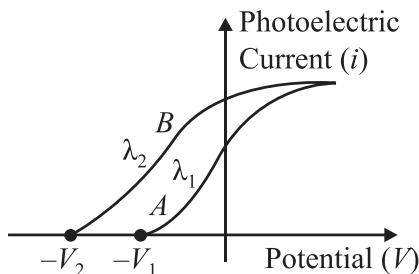
## SECTION-(I) PHYSICS

1. In photoelectric effect experiment, the graph between stopping potential and frequency for three different metals  $X$ ,  $Y$  and  $Z$  given below, the metal of maximum work function is:



- (1)  $X$  (2)  $Y$   
(3)  $Z$  (4) All of these

2. Select the correct statement for the following graph in a photoelectric experiment. Here  $\lambda_1$ ,  $\lambda_2$  are incident wavelengths,  $V_1$ ,  $V_2$  are stopping potentials.

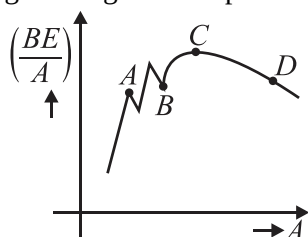


- (1) maximum kinetic energy of photoelectrons is greater in curve B than curve A.  
(2)  $\lambda_1 > \lambda_2$ .  
(3) frequency of incident radiation in curve B is greater than A.  
(4) all of these.

3. Which of the following can't be the angular momentum of an electron orbiting in a hydrogen atom?

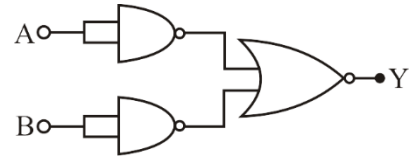
- (1)  $\frac{2h}{\pi}$  (2)  $\frac{h}{\pi}$   
(3)  $\frac{3h}{2\pi}$  (4)  $\frac{5h}{4\pi}$

4. The graph of binding energy per nucleon versus mass number  $A$  is drawn for nuclides as shown in figure.  $A$ ,  $B$ ,  $C$  and  $D$  are points corresponding to different elements. The element which is more likely to go through fission process is:



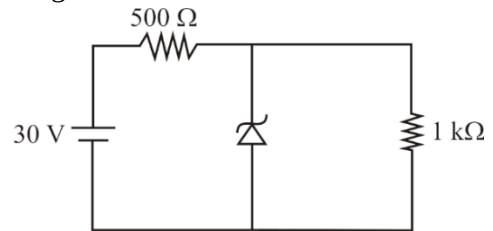
- (1)  $A$  (2)  $B$   
(3)  $C$  (4)  $D$

5. The operation performed by the circuit as shown in figure, is:



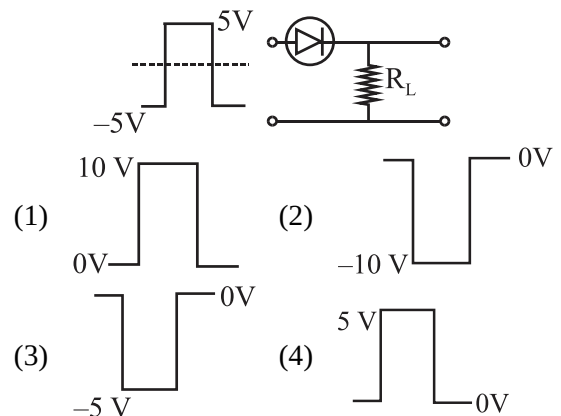
- (1) OR (2) NOR  
(3) AND (4) NAND

6. A zener diode of breakdown voltage  $V_z = 10\text{ V}$  is used in the circuit given below. The current through the zener diode is:



- (1) 40 mA (2) 30 mA  
(3) 20 mA (4) 10 mA

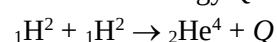
7. If in a P-N junction diode a square input signal of 10 V is applied as shown, then the output signal across  $R_L$  will be:



8. The ratio of the nuclear density of two nuclei of mass number 27 and 64 will be:

- (1) 3 : 4  
(2) 1 : 1  
(3)  $\sqrt{3} : \sqrt{4}$   
(4)  $(3)^{1/3} : (4)^{1/3}$

9. The binding energies per nucleon for a deuteron and an  $\alpha$  particle are  $x_1$  and  $x_2$  respectively. What will be the energy  $Q$  released in the reaction?

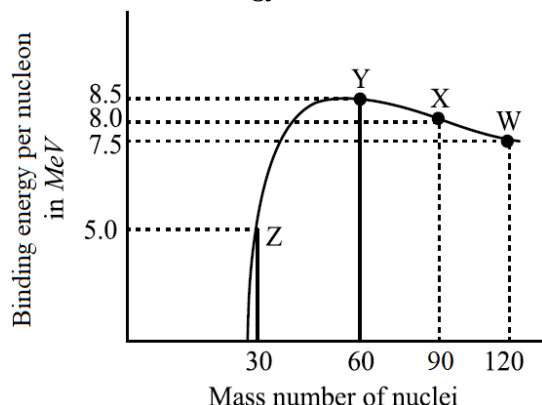


- (1)  $4(x_1 + x_2)$  (2)  $4(x_2 - x_1)$   
(3)  $2(x_1 + x_2)$  (4)  $2(x_2 - x_1)$

10. In the reaction  ${}^2_1\text{H} + {}^3_1\text{H} \rightarrow {}^4_2\text{He} + {}^1_0\text{n}$ , if the binding energies of  ${}^2_1\text{H}$ ,  ${}^3_1\text{H}$  and  ${}^4_2\text{He}$  are respectively  $a$ ,  $b$  and  $c$  (in MeV), then the energy (in MeV) released in this reaction is:

- (1)  $c + a - b$  (2)  $c - a - b$   
(3)  $a + b + c$  (4)  $a + b - c$

11. Binding energy per nucleon versus mass number curve for nuclei is shown in the figure. W, X, Y and Z are four nuclei indicated on the curve. The process that would release energy is:



- (1)  $Y \rightarrow 2Z$   
(2)  $W \rightarrow X + Z$   
(3)  $W \rightarrow 2Y$   
(4)  $X \rightarrow Y + Z$

12. Match List-I with List-II.

|     | List-I         |       | List-II   |
|-----|----------------|-------|-----------|
| (A) | Lyman series   | (I)   | $n_1 = 3$ |
| (B) | Balmer series  | (II)  | $n_1 = 2$ |
| (C) | Paschen series | (III) | $n_1 = 4$ |
| (D) | P fund series  | (IV)  | $n_1 = 1$ |
|     |                | (V)   | $n_1 = 5$ |

Choose the **correct** answer from the options given below:

- (1) A-IV, B-II, C-I, D-V  
(2) A-I, B-III, C-II, D-IV  
(3) A-II, B-IV, C-III, D-I  
(4) A-IV, B-I, C-II, D-III

[Short Practice Test-09 | Yakeen Plus (2025) | 17/11/2024]

Practical Organic Chemistry, Coordination Compound, The d and f-Block Elements

## SECTION-(II) CHEMISTRY

13. Ethylamine and diethylamine **cannot** be differentiated by

- (1) Hinsberg's test (2) Carbylamine test  
(3) Iodoform test (4) Both (1) and (2)

14. Test to differentiate between  $(\text{CH}_3\text{OH})$  and  $(\text{Ph} - \text{OH})$  is/are:

- (1) Litmus test (2)  $\text{FeCl}_3$   
(3)  $\text{Br}_2/\text{H}_2\text{O}$  (4) All of these

15. Match List-I with List-II.

|     | List-I<br>(Experiment)                                                   |       | List-II<br>(Inference)         |
|-----|--------------------------------------------------------------------------|-------|--------------------------------|
| (A) | The compound gives pink colour with Schiff's reagent.                    | (I)   | The compound is an acid.       |
| (B) | The compound gives offensive smell with chloroform and alcoholic KOH.    | (II)  | The compound is primary amine. |
| (C) | The compound gives effervescence with $\text{NaHCO}_3$                   | (III) | The compound is an aldehyde.   |
| (D) | The compound gives red colouration with ceric ammonium nitrate solution. | (IV)  | The compound is an alcohol.    |

Choose the **correct** answer from the options given below:

- (1) A-II, B-III, C-I, D-IV  
(2) A-II, B-III, C-IV, D-I  
(3) A-III, B-II, C-I, D-IV  
(4) A-III, B-II, C-IV, D-I

16. Give a simple test to differentiate cyclohexane and cyclohexene.

- (1)  $\text{Br}_2/\text{H}_2\text{O}$   
(2) Baeyer's Reagent  
(3) Tollen's reagent  
(4) Both (1) and (2)

17. Victor Meyer Test is used for the distinguished of  $1^\circ$ ,  $2^\circ$  and  $3^\circ$ :

- (1) Alcohols  
(2) Alkyl halides  
(3) Nitro Compounds  
(4) All of these

18. The no. of mole(s) of  $\text{AgCl}$  obtained when excess  $\text{AgNO}_3$  is added to one mole of  $[\text{Cr}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$ :

- (1) 1 (2) 2  
(3) 3 (4) 4

19. The IUPAC name of  $[\text{Pt}(\text{NH}_3)_3(\text{Br})(\text{NO}_2)(\text{Cl})]\text{Cl}$  is:
- (1) Triamminechloridobromidoplatinum(IV)chloride
  - (2) Triamminebromidochloridonitroplatinum(IV)chloride
  - (3) Triamminenitrochloribromidoplatinum(IV)chloride
  - (4) Triamminechloridonitrobromidoplatinum(IV)choride
20. Which of the following isomerism is **not** present in complex  $[\text{Co}(\text{NH}_3)_4(\text{SCN})_2]\text{Br}$ ?
- (1) Geometrical isomerism
  - (2) Linkage isomerism
  - (3) Ionisation isomerism
  - (4) Optical isomerism
21. Which of the following is **not** true for  $[\text{CoF}_6]^{3-}$ ?
- (1) It is an outer orbital complex.
  - (2) It is a high spin complex.
  - (3) The central metal ion has  $\text{sp}^3\text{d}^2$  hybridised orbitals.
  - (4) Co has +2 oxidation state and contains no unpaired electrons.

22. Which of the following pair of elements has  $(n-1)d^{10}ns^2$  electronic configuration?
- (1) Fe, Co, Ni
  - (2) Cu, Ag, Au
  - (3) Zn, Cd, Hg
  - (4) Sc, Y, La
23. The pair having similar magnetic moment is: (Atomic no.: Cr = 24, Mn = 25, Fe = 26, Ti = 22)
- (1)  $\text{Ti}^{3+}$ ,  $\text{V}^{3+}$
  - (2)  $\text{Cr}^{3+}$ ,  $\text{Mn}^{2+}$
  - (3)  $\text{Mn}^{2+}$ ,  $\text{Fe}^{3+}$
  - (4)  $\text{Fe}^{2+}$ ,  $\text{Mn}^{2+}$
24. The electronic configuration of gadolinium (At No. 64) is:
- (1)  $[\text{Xe}]4f^75d^16s^2$
  - (2)  $[\text{Xe}]4f^85d^06s^2$
  - (3)  $[\text{Xe}]4f^85d^16s^1$
  - (4)  $[\text{Xe}]4f^75d^06s^2$

[Short Practice Test-09 | Yakeen Plus (2025) | 17/11/2024]  
Organisms and Population, Ecosystem, Biodiversity and Conservation

## SECTION-(III) BOTANY

25. The formula for logistic growth is:
- (1)  $dN / dt = rN$ .
  - (2)  $rN / dN = dt$ .
  - (3)  $dN / dt = rN \left( \frac{K - N}{K} \right)$ .
  - (4)  $dN / dt = rN \left[ \frac{N - K}{N} \right]$ .
26. Which of the following are the characteristics of expanding population?
- A. Stable shaped age structure.
  - B. An urn shaped age structure.
  - C. Pre-reproductive and reproductive age groups become more or less equal in size.
  - D. Rapidly growing population with high birth rate.
- Choose the most appropriate answer from the options given below:
- (1) A and C only
  - (2) D only
  - (3) C and D only
  - (4) B and C only

27. Select the factors that will affect the primary productivity of an ecosystem.
- A. Soil moisture
  - B. Availability of nutrients
  - C. Plant species of that area
  - D. Photosynthetic capacity of plants.
- Choose the most appropriate answer from the options given below:
- (1) A is correct only
  - (2) B, C and D are incorrect
  - (3) All except C are correct
  - (4) A, B, C and D are correct

28. Match List-I with List-II.

| List-I |              | List-II |      |
|--------|--------------|---------|------|
| (A)    | Mutualism    | (I)     | +, + |
| (B)    | Parasitism   | (II)    | -, 0 |
| (C)    | Commensalism | (III)   | +, 0 |
| (D)    | Amensalism   | (IV)    | +, - |

Choose the **correct** answer from the options given below:

- (1) A-II, B-III, C-IV, D-I
- (2) A-I, B-IV, C-III, D-II
- (3) A-IV, B-II, C-I, D-III
- (4) A-III, B-IV, C-I, D-II

29. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:  
**Assertion A:** In an aquatic ecosystem producers include phytoplankton, algae and higher plants.

**Reason R:** Ecosystem need a constant supply of energy to synthesise the molecules they require, to counteract the universal tendency toward increasing disorderliness.

In the light of the above statements, choose the **correct** answer from the options given below.

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

30. Select the **correct** equation regarding GPP.

- (1)  $GPP - R = NPP$
- (2)  $GPP + R = NPP$
- (3)  $GPP/R = NPP$
- (4) All of these

31. Read the following and select the **correct** statement.

- (1) Warm and moist environment favour decomposition whereas low temperature and anaerobiosis inhibit decomposition.
- (2) Warm and moist environment inhibit decomposition whereas low temperature and anaerobiosis favour decomposition.
- (3) Warm moist environment and anaerobiosis inhibit decomposition whereas low temperature favours decomposition.
- (4) Warm and low temperature inhibit decomposition whereas anaerobiosis favours decomposition.

32. Select the **correct** option for **true** (T) and **false** (F).

- A. It has taken millions of years of evolution, to accumulate the rich diversity in nature, but we could lose all that wealth in less than two centuries if the present rates of species losses continue.
- B. According to the IUCN (2007), the total number of plant and animal species described so far is slightly more than 1.5 million.
- C. Robert May places the global species diversity at about 7 million.
- D. The biological wealth of our planet has been declining rapidly and the accusing finger is clearly pointing to human activities.

Choose the most appropriate answer from the options given below:

|     | A | B | C | D |
|-----|---|---|---|---|
| (1) | T | T | F | T |
| (2) | T | F | F | T |
| (3) | F | T | F | T |
| (4) | T | F | T | T |

33. Given below are two statements:

**Statement I:** The Western Ghats have a greater amphibian species diversity than the Eastern Ghats.

**Statement II:** Scandinavian country like Norway has a greater ecosystem diversity than India.

In the light of the above statements, choose the most appropriate answer from the options given below.

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

34. Although India has only \_\_\_\_ **P** \_\_\_\_ percent of the world's land area, its share of the global species diversity is an impressive \_\_\_\_ **Q** \_\_\_\_ percent. Identify **P** and **Q** respectively.

|     | P   | Q   |
|-----|-----|-----|
| (1) | 3.4 | 4.8 |
| (2) | 1.4 | 6.8 |
| (3) | 2.4 | 8.1 |
| (4) | 8.1 | 9.2 |

35. The highest number of species in the world from the following are represented by:

- (1) algae.
- (2) lichens.
- (3) fungi.
- (4) mosses.

36. Tiger subspecies that got extinct was/were:

- (1) Thylacine.
- (2) Quagga.
- (3) Caspian.
- (4) All of these

## SECTION-(IV) ZOOLOGY

37. Which of the following initiatives is a part of the Indian Patents Bill amendment?

- (1) Giving inadequate compensations.
- (2) Research and development initiatives.
- (3) Promote unauthorised exploitation of bio-resources.
- (4) Giving unequal benefit sharing.

38. Given below are two statements w.r.t Hardy Weinberg equilibrium.

**Statement I:** Allele frequencies in a population are stable but varies from generation to generation.

**Statement II:** Sum total of all the allelic frequencies is one.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

39. Which of the GMOs is/are used by biotechnology?

- (1) Microbes                      (2) Plants
- (3) Animals                      (4) All of these

40. What is main purpose of creating transgenic animals?

- (1) To enhance their physical appearance.
- (2) To study how genes affect the normal functions of the body and its development.
- (3) To increase their lifespan.
- (4) To improve their reproductive rates.

41. To insert foreign DNA into a vector, the vector should ideally have only one \_\_\_\_\_.

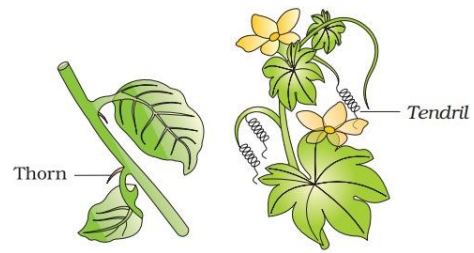
Choose the option that fills the blank **correctly**.

- (1) antibiotic resistance gene
- (2) recognition site
- (3) transformation site
- (4) ori

42. Which of the following statements is **correct** w.r.t primers?

- (1) It is a small chemically synthesised oligonucleotide.
- (2) It has similar sequence to the template DNA.
- (3) It is a small chemically synthesised polynucleotide.
- (4) Both (1) and (2)

43. The following given diagrams show;



- (1) convergent evolution in *Bougainvillea* and *Cucurbita*.
- (2) divergent evolution in *Bougainvillea* and *Cucurbita*.
- (3) divergent evolution in sweet potato and potato.
- (4) convergent evolution in sweet potato and potato.

44. Read the following statements (A-D).

- A. Restriction enzymes belong to a large class of enzymes called nucleases.
- B. *Hind II* cuts DNA at a specific sequence of seven base pairs.
- C. DNA molecules are positively charged.
- D. Restriction enzymes cut the sugar-phosphate backbone of DNA.

Which of the above given statements are **correct**?

- (1) A, B and D only
- (2) A and D only
- (3) A, C and D only
- (4) A, B, C and D

45. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:  
**Assertion A:** *Bt* cotton contains a gene from insect *Bacillus thuringiensis* that protects it from bacterial infection.

**Reason R:** The *Bt* toxin gene produces an insecticidal protein that kills insects upon ingestion by disrupting their gut cells.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

46. Select the **incorrect** option.

- (1) *Dryopithecus* was more man like.
- (2) *Homo habilis* probably did not eat meat.
- (3) *Homo erectus* had a cranial capacity of 900 cc.
- (4) Agriculture came around 10,000 years ago.

47. Read the following statements (A-E)
- A. Saltation is a single step large mutation.
  - B. The amphibians evolved into reptiles.
  - C. Branching descent and natural selection are two key concepts of Darwinian theory of evolution.
  - D. Sea weeds and few plants existed probably around 320 mya.
  - E. About 65 mya, the dinosaurs suddenly disappeared from the earth.

Which of the above statements are **correct**?

- (1) A, B and C only
- (2) A, D and E only
- (3) A, B and E only
- (4) A, B, C, D and E

48. Match **List-I** with **List-II**:

| List-I |              | List-II |                                                     |
|--------|--------------|---------|-----------------------------------------------------|
| (A)    | Pst I        | (I)     | Cleaves at specific position within the linear DNA. |
| (B)    | Endonuclease | (II)    | Amplification of genetic material                   |
| (C)    | Cellulase    | (III)   | amp <sup>R</sup>                                    |
| (D)    | PCR          | (IV)    | Degradation of plant cell wall                      |

Choose the **correct** answer from the options given below:

- (1) A-III, B-IV, C-I, D-II
- (2) A-IV, B-III, C-I, D-II
- (3) A-II, B-I, C-IV, D-III
- (4) A-III, B-I, C-IV, D-II

